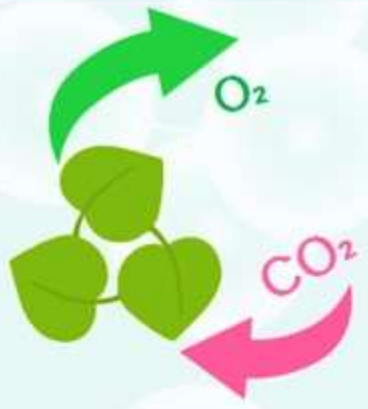


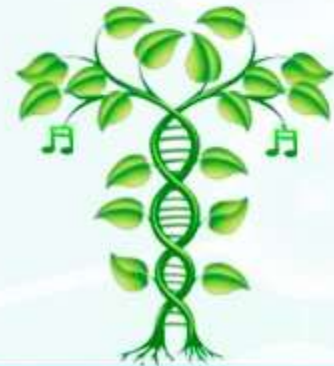


# BIOLOGY



## DEFICIENCY DISEASES

MODULE - 4



## DEFICIENCY DISEASE



### DEFICIENCY

A condition that occurs due to the unavailability of a particular nutrient over a long period of time. This types of diseases are non-contagious and can be reversed with a proper diet rich in the deficient nutrient.





# ENOUGH FOOD...STILL A LONG DREAM?



## CONDITION IN INDIA



- 1 India has a population of 1,344,774,227 people i.e. around 1.3 billion.
- 2 India has over 194.6 million of undernourished people who are incapable of eating a full balanced meal.
- 3 Even if one gets enough food to eat it may not contain a particular nutrient.
- 4 If this continues for a long period of time the person may suffer from its deficiency.
- 5 Nearly 276 million Indians are below the poverty level thus Deficiency disease is a **Major Issue** in India.



## DEFICIENCY OF CARBOHYDRATES



**MARASMUS**  
(Hindi - Shosha Rog)

### MARASMUS

- 1 Severe energy deficiency disease-lack of carbohydrates
- 2 Extreme thinning of body -  
**emaciation**
- 3 Stomach shrinkage and extreme dehydration
- 4 Muscular waste and loss of  
'under skin' body fat
- 5 Unpleasant odour in the breath





# MARASMUS - TREATMENT AND CONTROL



- 1 **Nutritional rehabilitation**- food given for 150 kilocalorie in a day
- 2 **Rehydration** by ORS and micronutrient supplementation like iron or vitamin A



## DEFICIENCY OF PROTEIN

### KWASHIORKOR



**KWASHIORKOR**  
*Hindi Sookha Rog*

- 1 Severe protein deficiency disease affecting young children
- 2 The child suffering may die before the age of five years
- 3 Symptoms include:
  - **Underweight child** with an pot belly due to enlarged liver
  - **Loss of appetite** and frequent diarrhea. The skin looks dark and scaly
  - **Oedema** or swelling of feet and face because water accumulates in tissues of feet and face.





## KWASHIORKOR - TREATMENT AND CONTROL



- 1
  - A protein rich diet should be given to the child.
  - Along with it mixed diet of wheat, gram, peanuts are useful
- 2 **Milk** should be in the diet mainly also including eggs, fish and pulses

***The disease originated in Ghana, Africa. Kwashiorkor means 'the sickness of a newborn' where mother's milk was not available to the baby which is their main source of protein***



## HOW TO POINT OUT THE EXACT DEFICIENCY?

### KWASHIORKOR

- 1 It affects children upto 5 years of age
- 2 Due to deficiency of **protein**
- 3 **Oedema** of legs and face
- 4 No degeneration of muscles



### MARASMUS

- 1 It affects children upto 1 year
- 2 Due to deficiency of **carbohydrate**
- 3 **No Oedema** is observed
- 4 Degeneration of muscles happen, body becomes very weak





# DEFICIENCY DISEASE DUE TO VITAMINS



## WHY ARE VITAMINS SO NECESSARY?

- 1 Vitamins are biomolecules needed for all the biochemical reactions happening in our body.
- 2 But, they cannot be produced by our body itself.
- 3 Vitamin B & C are soluble in water – they will simply flush out and we can't store it in our body.

***Vitamins are micronutrients - excess intake may lead to other conditions and even toxicity***



# VITAMIN A (RETINOL) DEFICIENCY



## NIGHT BLINDNESS

- **Rod cells** in the retina stop responding to light
- Rod cells contain **rhodopsin** protein which is synthesized by vitamin A
- Thus, in its absence, it becomes difficult or impossible to see in low light



## XEROPHTHALMIA

- **Dryness of the conjunctiva or cornea**
- It leads to corneal ulcers and total blindness in children

## PREVENTION





# VITAMIN B1 (THIAMINE) DEFICIENCY



## WET BERI BERI

- Affects the heart and circulatory system.
- Symptoms include shortness of breath and rapid heart rate.
- Lower limbs get swollen.



## DRY BERI BERI

- Affects the nerves.
- Muscles become weak in the limbs.
- Can lead to paralysis.

## PREVENTION



# VITAMIN B2 (RIBOFLAVINE) DEFICIENCY



## ARIBOFLAVINOSIS

- Dryness around areas of lips and nose
- Symptoms include a sore throat and inflamed tongue
- It is also caused when the liver is not functioning properly (sometimes in alcoholics)
- Protein rich foods should be taken which also contain Vit. B2

## PREVENTION





# VITAMIN B3 (NIACIN) DEFICIENCY



## PRIMARY PELLAGRA

- It occurs when the diet doesn't contain enough niacin
- Symptoms include peeling and redness of skin
- Other symptoms are same as secondary pellagra



## SECONDARY PELLAGRA

- It occurs due to poor ability of the body to use niacin from diet
- It can result due to prolonged alcoholism or long term diarrhea
- The symptoms are diarrhea, dermatitis, dementia or forgetfulness and death.

## PREVENTION



## VITAMIN B5 (PANTOTHENIC ACID) DEFICIENCY



- Vitamin B5 deficiency is rare
- It can show symptoms like **fatigue**, loss of coordination and **irritation**
- Deficiency can also result in vomiting and stomach pain

### PREVENTION





# VITAMIN B6 (PYRIDOXINE) DEFICIENCY



## NERVE DAMAGE

- Vitamin B6 is also called Brain vitamin along with Vit. B9 and Vit. B12
- Deficiency of pyridoxine results in **nerve damage** which may impair sensation in the limbs
- Skin and scalp becomes **flaky and itchy**

## PREVENTION



# VITAMIN B7 (BIOTIN) DEFICIENCY



## LETHARGY

- Biotin deficiency can be even fatal if left untreated.
- Symptoms include decreased appetite and lethargy.
- Thinning of hair, hair loss and permanent baldness.
- Skin and scalp becomes flaky and itchy with fungal infections.

## PREVENTION





# VITAMIN B9 (FOLATE/FOLIC ACID) DEFICIENCY



## ANEMIA

- Body has **very few RBC** and the cells are deprived of oxygen.
- Folate deficiency in pregnant woman can cause severe even fatal **birth defects** of the child.
- Poor growth, forgetfulness and mental sluggishness is observed in children.

## PREVENTION



# VITAMIN B12 (COBALAMINE) DEFICIENCY



## PERNICIOUS ANEMIA

- Body cannot make adequate healthy RBC.
- Defect in the DNA synthesis of RBC in blood – Megaloblastic Anemia
- Mild diarrhea and difference in bowel movement
- Degeneration of spinal cord which results in dementia
- In infants it can cause severe neuronal damage

## PREVENTION





# VITAMIN C - DEFICIENCY

## SCURVY

- Early symptoms are discomfort and lethargy progressing to shortness of breath and bone pain.
- Skin turns rough with recurring hair fall and increase of the girth.
- The gums swell and bleed and if left untreated can loosen around the teeth causing early tooth loss.



## PREVENTION



# VITAMIN D DEFICIENCY

## BONE DEMINERALISATION

- The bones becomes soft and undergo osteoporosis.
- Osteomalacia a bone thinning disorder in adults and increased bone fragility.
- Rickets in children with deformity in long bones and thinning of skull.
- Low levels of vit. D also increases depression in patients.



## PREVENTION





# VITAMIN E DEFICIENCY

## NEUROLOGIC DEFICITS & RETINOPATHY

- It causes nerve problems due to poor conduction of electrical impulses along nerves.
- In children it causes neurologic deficits, including muscle weakness and spinocerebellar ataxia with loss of deep tendon reflexes.
- Vitamin E deficiency may contribute to retinopathy or retinal damage in premature infants and to some cases of intraventricular hemorrhage in neonates.



## PREVENTION



# VITAMIN K DEFICIENCY

## HEMORRHAGE

- Blood continues to ooze at surgical or puncture sites, risk of massive uncontrolled bleeding.
- Stomach pains and cartilage calcification; and severe malformation of developing bone or deposition of insoluble calcium salts in the walls of arteries.
- In infants, it can cause some birth defects such as underdeveloped face, nose, bones, and fingers.



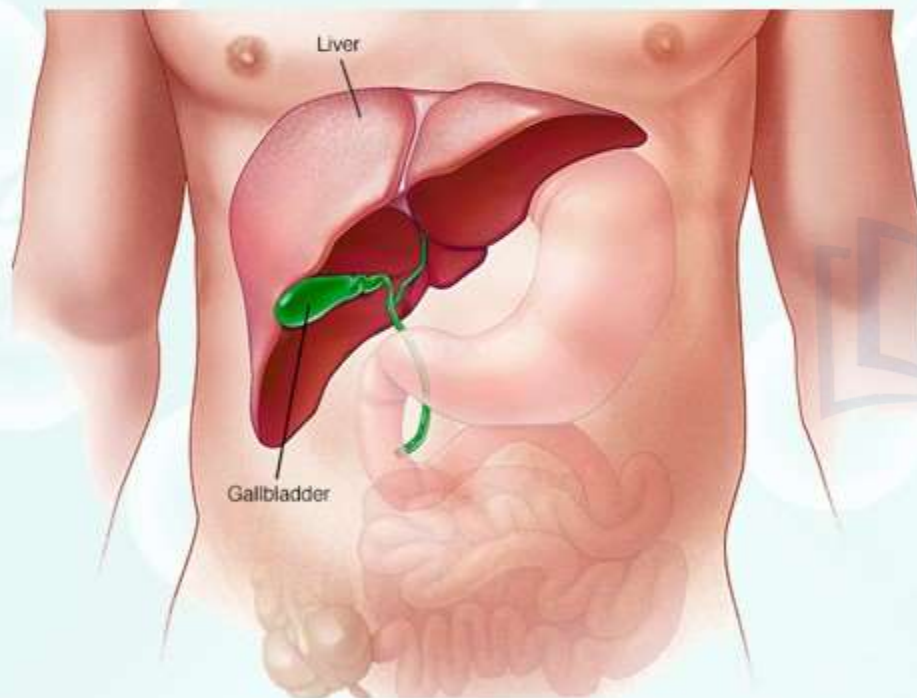
## PREVENTION





# WHAT IF VITAMINS ARE OVERDOSED?

## HYPERVITAMINOSIS



- Vitamin supplements can be harmful when taken in excessive amounts.
- Fat soluble Vitamins **A, D, E & K** can stay in our body for a long time in liver or fatty tissues under the skin.
- Vitamin overdose can affect the **liver** and can even damage it.
- Vitamins can build up in the body and can cause fatal illness.



## EFFECTS OF COMMON VITAMIN OVERDOSE

### HYPERVITAMINOSIS

HYPERVITAMINOSIS A

HYPERVITAMINOSIS D

HYPERVITAMINOSIS E

HYPERVITAMINOSIS K

HYPERVITAMINOSIS B3

HYPERVITAMINOSIS B6

### EFFECTS

Bone becomes soft, liver damage

High levels of calcium in blood, bones become soft

Increased bruising & bleeding, weakness

Increased blood clot formation

Damage liver

Damage nerves, difficulty while walking





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## OUR TEAM

G. V. Rao

Amrita Naidu

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

K. Srikanth

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